

Current Mirror: Simple and Cascode

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Programme	B.Tech Electronics and Communication Engineering
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Tools	Cadence Virtuoso

1. Aim

To design and simulate a simple NMOS current mirror and a cascode NMOS current mirror in Cadence Virtuoso, and to compare their output (I–V) characteristics.

2. Design Specifications

2.1 Simple Current Mirror

Parameter	Value
Device	nmos1v (gpdk090 nmos1v)
Width (W)	100 μm
Length (L)	200 nm
Reference current (I ₂)	1 mA
Reference branch supply (V1)	1.2 V
Technology	GPDK090

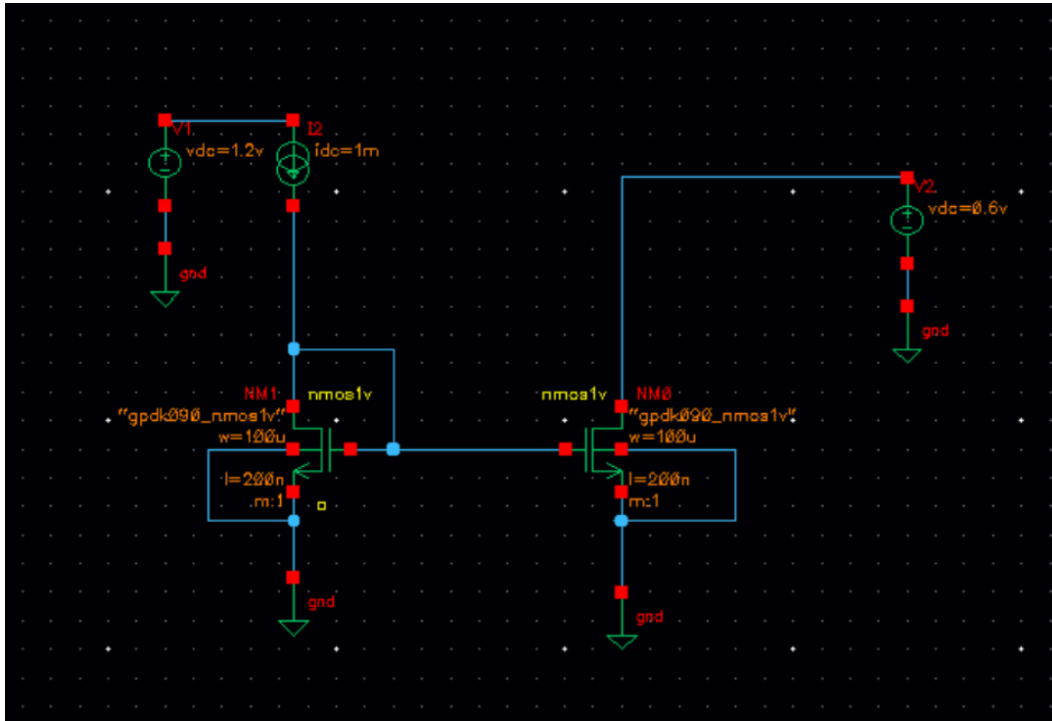
2.2 Cascode Current Mirror

Parameter	Value
Device	nmos1v (gpdk090 nmos1v)
Width (W)	200 μm
Length (L)	200 nm
Reference current (I ₁)	2 mA
Reference branch supply (V0)	1.2 V
Technology	GPDK090

3. Circuit

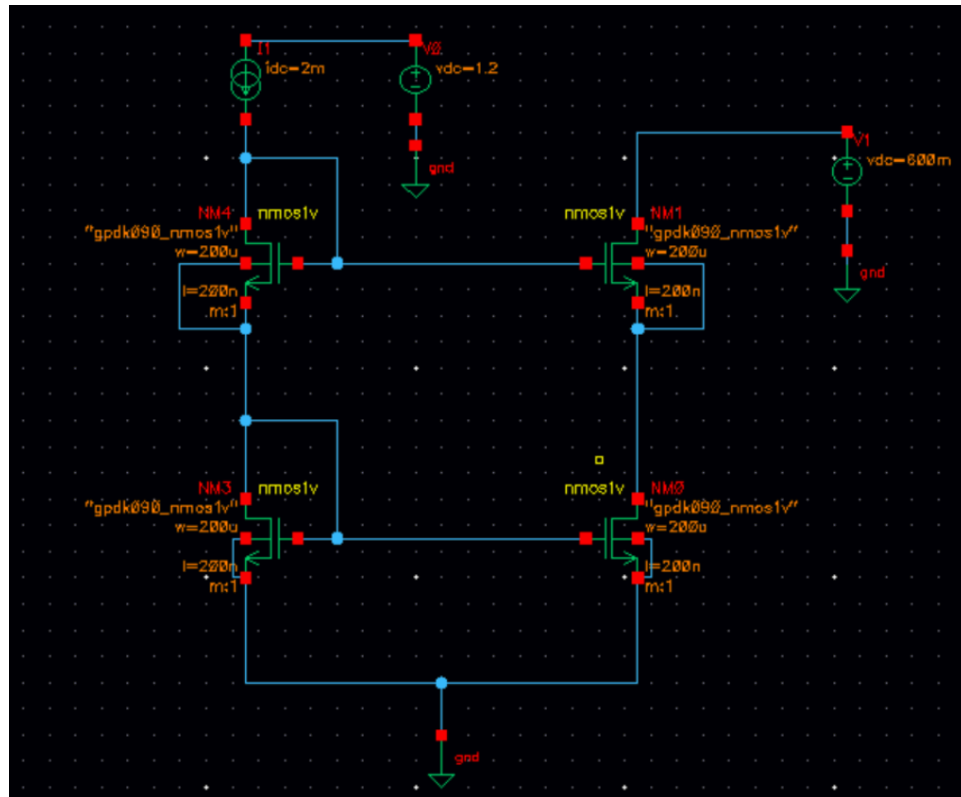
3.1 Simple Current Mirror

Two matched `nmos1v` devices, NM1 and NM0, form the mirror. NM1 is diode-connected (gate tied to drain) and carries the 1 mA reference current supplied by an ideal current source (I2) from a 1.2 V rail (V1). The resulting gate-source voltage is applied to NM0, whose drain forms the output node. An independent DC source (V2) biases the output node and is swept to trace the mirror's output characteristic.



3.2 Cascode Current Mirror

The cascode mirror stacks a second diode-connected device above each transistor of the simple mirror to raise the output resistance. NM3 (bottom, reference side) is diode-connected and carries the 2 mA reference current from I1; its gate also biases the output-side mirror device NM0 (bottom). NM4 (top, reference side) is diode-connected in series above NM3 and biases the cascode device NM1 (top), which is stacked above NM0 on the output side. The output node, at the drain of NM1, is biased by an independent DC source (V1) swept from 0 to 1.2 V.



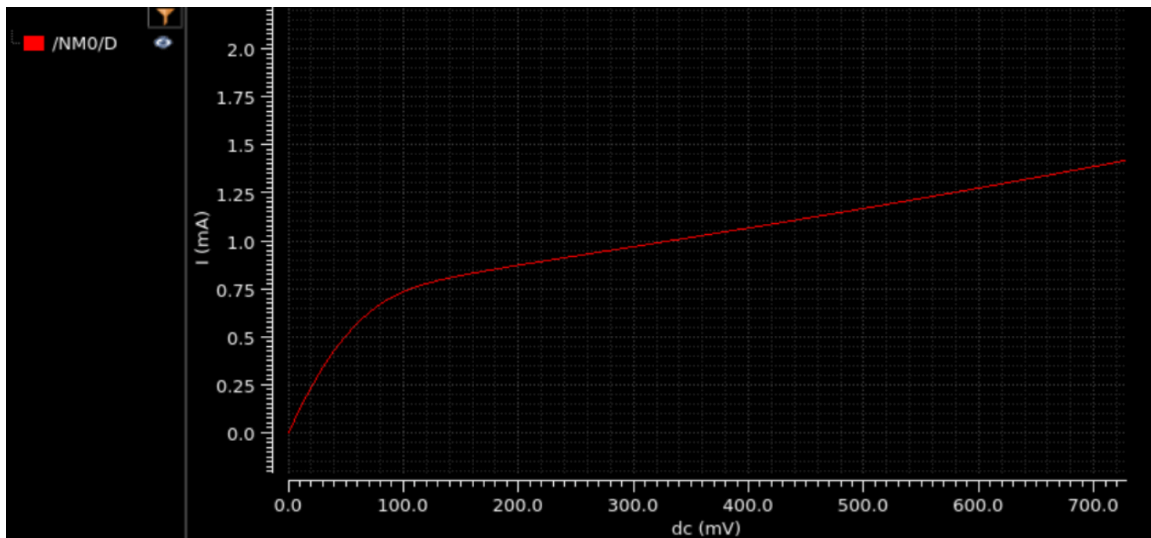
4. Simulation Procedure

- Drew the simple two-transistor mirror in Virtuoso with a 1 mA ideal reference current source (I2) on the diode-connected side and an independent DC source (V2) on the output side.
- Swept V2 (output node voltage) and plotted the drain current of NM0 to obtain the simple mirror's output I-V characteristic.
- Drew the four-transistor cascode mirror, stacking a diode-connected cascode device above each transistor of the simple mirror, with a 2 mA ideal reference current source (I1).
- Swept V1 (output node voltage) from 0 to 1.2 V and plotted the drain current of NM1 to obtain the cascode mirror's output I-V characteristic.

5. Results

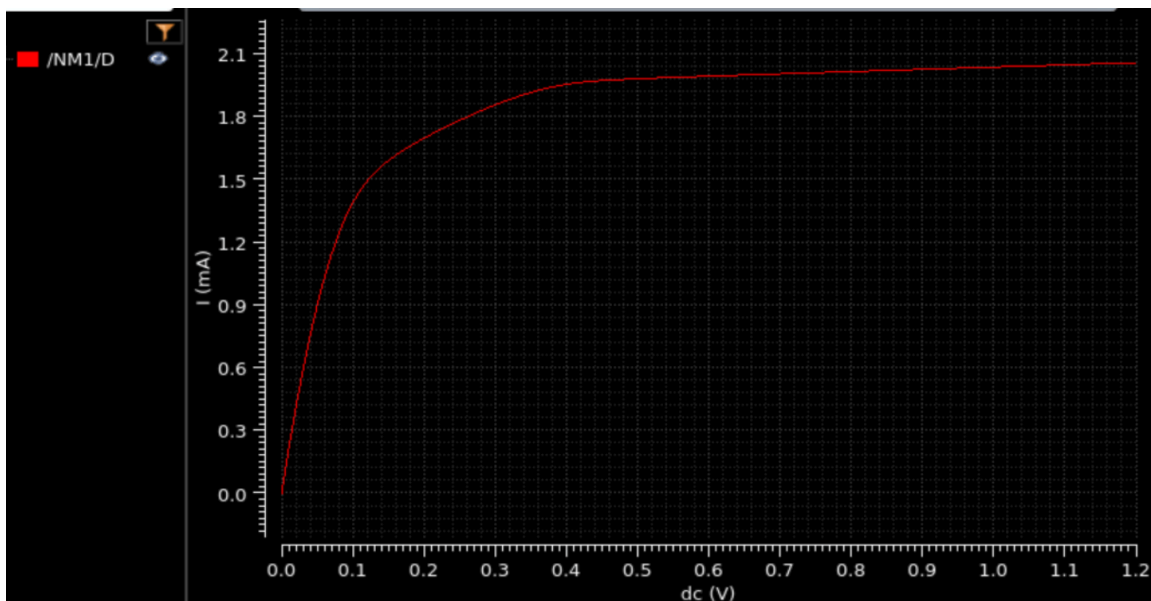
5.1 Simple Current Mirror — Output Characteristic

The mirrored current rises sharply as NM0 leaves the triode region and approaches the 1 mA reference current, but keeps increasing



5.2 Cascode Current Mirror — Output Characteristic

The mirrored current rises quickly and settles near the 2 mA reference current by about 0.4 V, remaining flat



6. Observations

- Both mirrors reproduce their respective reference currents (1 mA and 2 mA) once the output device enters saturation.
- The simple mirror's output current continues to rise across the swept output voltage range, indicating a relatively low output resistance.
- The cascode mirror's output current flattens out well before the sweep ends, showing a clearly higher output resistance than the simple mirror.
- The improved output resistance of the cascode mirror comes at the cost of a higher minimum output (compliance) voltage before the mirrored current settles, compared to the simple mirror.

7. Conclusion

A simple and cascode current mirror were designed and simulated in Cadence Virtuoso. Both mirrors accurately reproduced their respective reference currents, and the cascode mirror showed a visibly flatter output characteristic than the simple mirror, confirming that cascoding significantly increases the output resistance of a current mirror at the cost of reduced output voltage.